

“Hands-on” Light Field Display via Engineering Scalable Optics, Algorithms, and Mounts

Feifan Qu

University of Hong Kong
Hong Kong, Hong Kong
feifan.qu@connect.hku.hk

Li Liao

University of Hong Kong
Hong Kong, Hong Kong
liliao24@connect.hku.hk

Pengfei Shen

University of Hong Kong
Hong Kong, Hong Kong
pengfei.shen@connect.hku.hk

Yifan Peng

University of Hong Kong
Kong Hong, Hong Kong
evan.y.peng@gmail.com

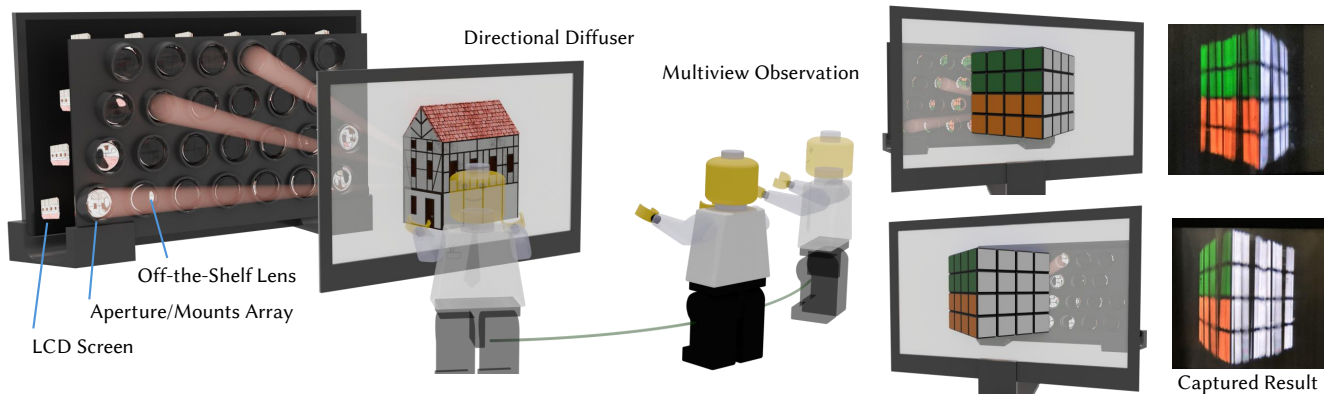


Figure 1: Overview of the “hands-on” light field display, illustrating the system architecture and resulting display effect. (Left) Conceptual illustration of the light field display system, built upon off-the-shelf parts. A dense array of mini-projectors emits patterned light through a corresponding lens array. A directional diffuser screen then integrates these incoming rays to synthesize the 3D light field within the viewing zone. This architecture enables the shared, glasses-free 3D experience for multiple, freely-moving observers. (Right) Demonstration of view-dependent display. The simulation on the left illustrates a target scene rendered from two distinct viewpoints. On the right, photographs captured directly from our hardware prototype showcase the resulting parallax effect, with the cube’s perspective shifting convincingly as the viewing position changes.

Abstract

We present a scalable, “hands-on” light field display for shared, glasses-free Extended Reality (XR). Our system uses an array of mini-projectors, built from low-cost, off-the-shelf components, to project a 3D light field onto a directional diffuser screen. This provides multiple viewers with vivid scenes featuring continuous motion parallax and correct stereo parallax. Our core technical contributions are a systematic hardware and software pipeline. Hardware precision is bootstrapped using 3D-printed mounts and finalized with a camera-assisted calibration method that compensates for optical and geometric errors. A physically-based ray tracing algorithm ensures the final imagery is realistic. The integration of gesture

tracking enables intuitive interaction, demonstrating a practical and accessible approach to building collaborative XR systems.

Keywords

Light field display, extended reality, rendering, calibration

ACM Reference Format:

Feifan Qu, Pengfei Shen, Li Liao, and Yifan Peng. 2025. “Hands-on” Light Field Display via Engineering Scalable Optics, Algorithms, and Mounts. In *SIGGRAPH Asia 2025 XR (SA XR ’25)*, December 15–18, 2025, Hong Kong, Hong Kong. ACM, New York, NY, USA, 3 pages. <https://doi.org/10.1145/3761667.3761957>

1 Introduction

The pursuit of shared Extended Reality (XR) demands displays that can present vivid 3D imagery to multiple, free-moving observers without heavy headgear. While head-mounted displays (HMDs) excel in single-user immersion, they inherently isolate users from one another [Ebner et al. 2022; Kouliris et al. 2017]. Conventional glasses-free solutions, such as autostereoscopic screens, attempt to bridge this gap but are often constrained by a discrete set of views,

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

SA XR ’25, Hong Kong, Hong Kong

© 2025 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-2168-7/25/12

<https://doi.org/10.1145/3761667.3761957>

breaking the crucial illusion of motion parallax. Light field reconstruction offers an ideal solution, providing the smooth, continuous parallax necessary for depth intuition.

However, transitioning from principle to a practical light field display is challenging due to high costs, complex optics, and demanding calibration, which have limited widespread adoption [Peng et al. 2012; Yang et al. 2007; Yasui et al. 2024]. Our work confronts these barriers with a systematic engineering approach, integrating scalable off-the-shelf optics, high-fidelity rendering, and precision mechanical mounting. This creates a powerful yet accessible platform, demonstrating a viable pathway toward collaborative XR in fields like medical visualization, art design, and education.

2 Method and Implementation

Our system is engineered with a *modular* philosophy, allowing standalone display units to be tiled for expanded, large-scale experiences. The foundation of this “hands-on” solution is an end-to-end pipeline that combines low-cost, off-the-shelf optics with precision manufacturing and advanced computational algorithms.

The optical core of our prototype features a 17.3-inch 4K display partitioned by a lens and aperture array into a grid of “micro-projector units.” This design is highly scalable, as our parametric CAD pipeline automatically generates the necessary 3D-printable mounts to accommodate different screen sizes. These micro-projectors project unique sub-images onto a directional diffuser screen, which then fuses the discrete projections into a continuous and seamless light field for the viewer through its anisotropic scattering properties.

We ensure precision through a two-stage process combining physical fabrication and computational correction. First, customized structural components, such as lens mounts, are 3D-printed to enable a high degree of initial alignment. Then, a camera-based feedback compensates for residual optical aberrations and misalignments. This system captures test patterns from multiple views to generate a pre-warp map. Our tailored ray tracing renderer applies this correction map to synthesize a physically realistic light field with global illumination. This complete pipeline enables high-fidelity results from accessible, low-cost parts.

3 Results and Discussion

Our work has resulted in a functional prototype for a compelling, shared XR experience. Built from a single commodity LCD, the system generates a floating 3D scene with a 30° horizontal field of view, showing that high-quality 3D visualization is feasible using accessible components. The prototype is well-suited for desktop inspection and interaction. Its primary hardware limitation relates to the 3D-printed fabrication. Our printer’s building volume limited us to create large components in segments for later assembly. This process can introduce minor alignment errors, occasionally creating thin, dark seams in the final image. Manufacturing these parts as a single piece on a larger printer would mitigate this artifact.

As shown in Figure 2, the displayed models appear vivid and spatially stable, with correct occlusion and physically plausible lighting. Observers can move around the display and experience smooth motion parallax. This freedom of movement is essential for

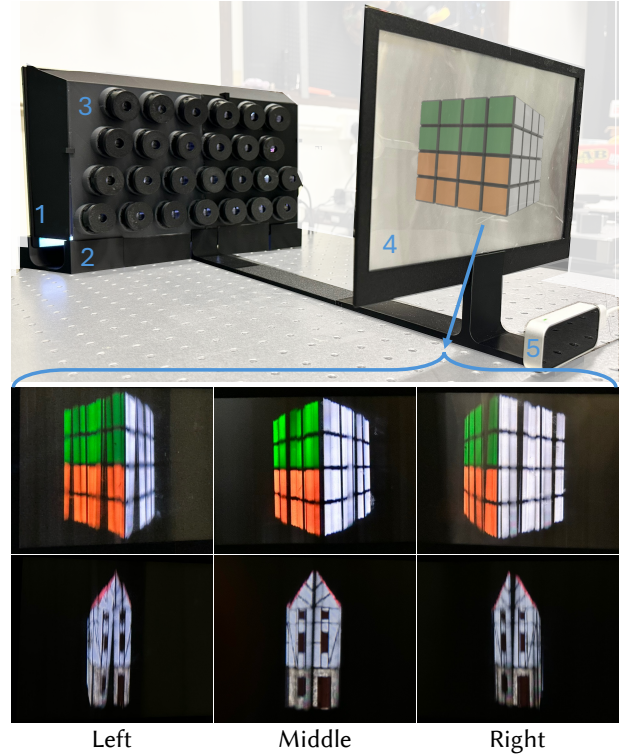


Figure 2: Experimental prototype (Top) and captured view-dependent result (Bottom). The fully assembled prototype consists of (1) a 4K LCD screen that provides the source imagery, housed in (2) a customized, 3D-printed frame which ensures precise alignment for (3) the mini-projector lens array. The light field is then formed on (4) a directional diffuser screen. For intuitive interaction with the 3D content, (5) an off-the-shelf gesture sensor is integrated. The display results of 3D models (cube and house) acquired from varying viewpoints reveal smooth motion parallax and occlusion.

a natural experience, allowing multiple users to inspect a virtual object from their own unique perspectives for collaborative tasks.

In summary, our “hands-on” display demonstrates a complete system that successfully integrates scalable optics, advanced rendering, and precision mounts. This synergy creates a powerful and accessible platform for interactive, glasses-free XR applications. The experience is made fully tangible through an integrated gesture-tracking module, which allows users to intuitively manipulate virtual objects. While the current prototype has limitations, such as display brightness and minor image stitching artifacts, we are confident these can be addressed in future work through backlight modification and further refinements to the mechanical design.

Acknowledgments

This work was supported by the National Natural Science Foundation of China (62322217) and the Research Grants Council of Hong Kong (ECS 27212822, GRF 17208023).

References

- Christoph Ebner, Shohei Mori, Peter Mohr, Yifan Peng, Dieter Schmalstieg, Gordon Wetzstein, and Denis Kalkofen. 2022. Video see-through mixed reality with focus cues. *IEEE Transactions on Visualization and Computer Graphics* 28, 5 (2022), 2256–2266.
- George-Alex Koulieris, Bee Bui, Martin S. Banks, and George Drettakis. 2017. Accommodation and comfort in head-mounted displays. *ACM Trans. Graph.* 36, 4, Article 87 (July 2017), 11 pages. doi:10.1145/3072959.3073622
- Yifan Peng, Haifeng Li, Rui Wang, Qing Zhong, Xiang Han, Zisheng Cao, and Xu Liu. 2012. Self-calibration three-dimensional light field display based on scalable multi-LCDs. *Journal of the Society for Information Display* 20, 12 (2012), 653–660.
- Ruigang Yang, Xinyu Huang, Sifang Li, and Christopher Jaynes. 2007. Toward the light field display: Autostereoscopic rendering via a cluster of projectors. *IEEE Transactions on Visualization and Computer Graphics* 14, 1 (2007), 84–96.
- Masahiko Yasui, Ryota Iwataki, Masatoshi Ishikawa, and Yoshihiro Watanabe. 2024. Projection Mapping with a Brightly Lit Surrounding Using a Mixed Light Field Approach. *IEEE Transactions on Visualization and Computer Graphics* (2024).